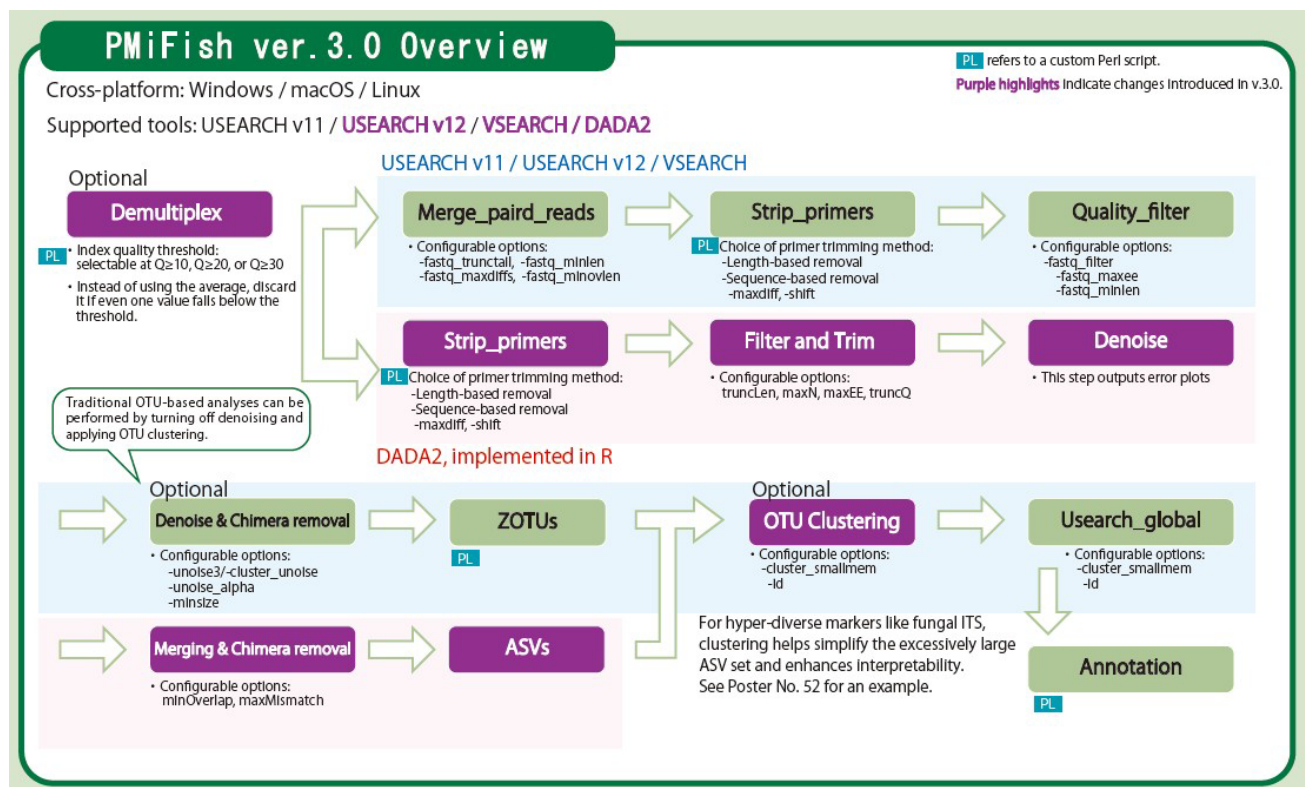


PMiFish (A Pipeline for MiFish Data Analysis) User's manual

Version 3.0 (2026/6/24)

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PMiFish is a streamlined pipeline designed for the efficient analysis of environmental DNA (eDNA) metabarcoding data. By taking FASTQ files generated from Illumina next-generation sequencers as input, the pipeline performs a comprehensive workflow through to taxonomic assignment. The core analysis relies on **USEARCH** (Edgar, 2010) or **VSEARCH** (Rognes et al., 2016), with the additional option to use **DADA2** (Callahan et al., 2016) for denoising. A key advantage of this pipeline is its user-friendliness; once the initial configuration is complete, subsequent analyses can be executed with minimal effort. Despite being commonly referred to as the "**MiFish Pipeline**," it is highly versatile and can be applied to eDNA metabarcoding for any taxonomic group, not just fish. Furthermore, it offers high analytical flexibility by allowing users to process samples involving multiple primer sets, generating individual analysis results for each primer used.



1. Environment Setup

【Settings Required for Windows Only】

- Install Strawberry Perl
<https://strawberryperl.com/>
- Install gzip
<http://gnuwin32.sourceforge.net/packages/gzip.htm>

Note: gzip does *not* automatically add its directory to the system PATH.

You must add it manually:

1. System Properties → Advanced system settings
2. Under the Advanced tab, click Environment Variables
3. Select the Path variable and click Edit
4. Add the directory path containing gzip.exe at the end of the list
Example: C:¥Program Files (x86)¥GnuWin32¥bin

【Common Settings (Windows, macOS, Linux)】

PMiFish can run using either USEARCH or VSEARCH.

Install one of them before running the pipeline.

If you wish to use DADA2 for denoising, you must also install R and the DADA2 library.

Using USEARCH

- Download the version appropriate for your OS from:
<https://www.drive5.com/usearch/>
- Place the executable into the Tools directory.
(Both v11 and v12 are available, but I recommend v11 as v12 has a more simplified feature set.)

Using VSEARCH

- Download the latest release for your OS from:
<https://github.com/torognes/vsearch/releases/>
- Extract the downloaded file and copy the contents of the bin directory into the Tools directory.

Using DADA2

1. Install R

Download and install the latest version from:

<https://www.r-project.org/>

2. Install the DADA2 library

Reference: <https://benjjneb.github.io/dada2/dada-installation.html>

In R, execute the following commands

(Replace v1.16 with the latest version if necessary):

```
r
install.packages("devtools")
library(devtools)
devtools::install_github("benjjneb/dada2", ref="v1.16")
```

When using Swarm (Mahé et al., 2022) for clustering:

- Download the version of **Swarm** (<https://github.com/torognes/swarm/releases>) compatible with your OS and place it in the **Tools** directory.

When performing post-clustering curation with LULU (Frøslev et al., 2017):

- **R** must be installed, along with the **LULU** library.

2. Directory Structure

PMiFish/

- |— DataBase/ # Reference DB (FASTA) and primer file
- |— Dictionary/ # Species/common name dictionaries etc.
- |— Results/ # All output files
- |— Run/ # Input FASTQ or FASTQ.gz
- |— Scripts/ # Perl scripts for each step
- |— Tools/ # USEARCH / VSEARCH executables
- |— PMiFish.pl # Main pipeline script
- |— Demultiplex.pl # Script for demultiplex
- |— PA_with_DB.pl # Script for phylogenetic analysis with database
- |— Setting.txt # User settings
- |— Options_DADA2.tx
- |— Options_usearch.txt

3 How to Run the Analysis

STEP 1 — Configure the settings

Edit Setting.txt to specify the required parameters.

If the same settings are reused, this step does not need to be repeated each time.

If you need to change the default values for USEARCH/VSEARCH or DADA2, edit:

- Options_usearch.txt
- Options_DADA2.txt

STEP 2 — Prepare input files

Place the FASTQ or FASTQ.gz files to be analyzed into the Run directory.

STEP 3 — Execute the pipeline

Open a command-line interface in the directory where PMiFish.pl is located:

- Windows: Open PowerShell
(Shift + Right-click inside the directory → “Open PowerShell window here”)
- macOS / Linux: Open Terminal and navigate to the directory

Run the pipeline using:

```
perl ./PMiFish.pl
```

If you wish to run steps separately, execute:

```
perl ./Scripts/_____.pl
```

If you encounter an error such as "**Can't spawn ...**" on Windows, refer to *Appendix 7*.

If macOS/Linux shows "**permission denied**", move to the **Tools** directory and run:

```
chmod +x <USEARCH_filename>
```

STEP 4 — Check the output

All results will be saved in the Results directory

4 Output Files and Directories

- **Note on Setting.txt:**
 - If **Temporary = YES**, FASTQ files from steps 1-1 to 1-3 will be automatically deleted.
(As these files are large, it is generally recommended to set this to YES).

- If **Compress = YES**, FASTQ files will be compressed into GZ format. (Note: Compression takes time. Also, **USEARCH cannot process files while they are compressed**).

◆ When using USEARCH/VSEARCH for Denoising

- **1-1 Merge_paired_reads**: Merged R1 and R2 reads and their log files.
- **1-2 Strip_primers**: Reads after primer removal and log files.
- **1-3 Quality_filter**: Reads after quality control (QC) and log files.
- **2-1 Find_unique_in_each_sample**: Dereplicated files containing unique sequences per sample.
- **2-2 Denoise**: Checking for PCR errors and chimeric sequences.
 - `_unoise3_result.txt`: Results of the denoising process.
 - `_zotu.fa`: Sequences with PCR errors removed (still contains chimeras).
- **2-3 Separate_chimera**: Sequences separated into chimeras and non-chimeras.
 - `_zotu_chimeras.fa`: Chimeric sequences.
 - `_zotu_nonchimeras.fa`: Non-chimeric sequences.
- **2-4 ZOTUs**: Combined ZOTU (Zero-radius OTU) files for all samples.

◆ When using DADA2 for Denoising

- **1-1 Primer_Trimmed_fastq**: Reads after primer removal and log files.
- **1-2 Filtered**: Reads after quality filtering.
- **2-4 ASVs**: Combined ASV (Amplicon Sequence Variant) files for all samples.

◆ Common Downstream Analysis

- **2-5 Rarefaction (optional)**: Files after rarefaction processing.
- **2-6 OUT_Clustering (optional)**: Files after clustering.
- **2-7 LULU (optional)**: Files after post-clustering curation using LULU.
- **3-1 Usearch_global**: Results from the `usearch_global` command.
- **4-1 Annotation**: A set of files summarizing the annotation results.
 - `_all_annotated_seq.fas`: All unique sequences from the samples.
 - `_Detail.html`: Detailed report of the results.
 - `_Representative_seq.fas`: A collection of representative sequences.
 - `_Summary.txt`: Executive summary of the findings.
 - `_neighbor_list.html`: List of detected species meeting the defined identity threshold.

- **5-1 Fasta_for_Phylogenetic_Analysis:** Representative sequences prepared for phylogenetic analysis.
 - `all_representative_seqs.fas`: Consolidated representative sequences from all samples.
 - `merged_list.txt`: List created when identical sequences are merged.
 - `merged_seq.fas`: Unique sequences from `all_representative_seqs.fas` after merging duplicates.
 - `merged_seq_with_family_name.fas`: Merged sequences with Family names appended.
- **5-2 Summary_Table:** Tables summarizing results by taxonomic group and representative sequences.
- **5-3 Fasta_classified_by_Family:** Fasta files organized by Family.
 - `list.txt`: List showing the number of species contained in each file.
 - `__.fas`: Fasta files per Family (`N.A.fas` contains sequences with unknown Family).
- **Portal_[Date_Time].html:** Comprehensive portal site summarizing all results.

◆ Logs and Configuration Records

- **log.html:** HTML file summarizing read counts after each processing step.
- **Setting_log.txt:** Copy of the `Setting.txt` used for the analysis.
- **Options_usearch_log.txt:** Copy of the `Options_usearch.txt` used.
- **Options_DADA2_log.txt:** Copy of the `Options_DADA2.txt` used.

5 Update History

■ Version 3.0 (2026/5/7)

[Major Changes]

- **Expanded Tool Support:** Added compatibility for **USEARCH v12** and **VSEARCH**.
- **Added DADA2 Support:** Users can now select **DADA2** for the denoising process.
- **New Script: 2_5_ASVs.pl**
 - Pools reads from all samples and outputs ASV/ZOTU/OTU tables and sequences.
- **New Script: 2_6_Clustering.pl**
 - Adds an option to perform clustering for ASVs/ZOTUs/OTUs.
 - Allows selection between **USEARCH/VSEARCH** and **SWARM**.

- **New Script: 2_7_LULU.pl**
 - Adds an option for **Post-Clustering curation** using LULU.
- **Output Method for Multiple Primers:** Modified the output format for multi-primer analyses.
 - Individual results for each primer are now output in 3_1_USEARCH_global, 5_2_Summary_Table, and 5_3_Fasta_classified_by_Family.
- **Enhanced Configuration:**
 - Added **Options_USEARCH.txt**: Clearly defines USEARCH/VSEARCH options used in each step and allows users to modify their values.
 - Added **Options_DADA2.txt**: Clearly defines DADA2 options and allows modification.
- **New Features & Scripts:**
 - Added **LCA (Lowest Common Ancestor) Rank** to Portal.html and Summary_Table.tsv.
 - Added **Demultiplex.pl** for demultiplexing tasks.
 - Added a suite of scripts for easier creation of custom reference databases from GenBank format: **GB2DB.pl**, **Amplified_region.pl**, **Cluster_maker.pl**, and **Taxonomic_Rank.pl**.

[Minor Changes]

- Renamed 2_1_Find_Unique to **2_1_Find_unique_in_each_sample**.
- Updated 3_1_Usearch_global to output Taxonomy_assignment_table.tsv, Taxonomy_assignment.fa, nohit.fa, and nohit_list.txt.
- Removed 6_2_Phylogenetic_trees.
- Fixed a bug where samples with zero reads were not displayed in the Portal (USEARCH/VSEARCH only).
- Renamed Portal.html to **Portal_[Date_Time].html**.
- Added an **Analysis Overview** section to Portal.html.
- Updated the tables in Portal.html to display read counts and their percentages for each item.
- The Results directory now includes log files for **Setting** and **Options**.
- Revised the format of **Setting.txt**.

■ Version 2.4 (2018/12/15)

- Updated 1_2_Strip_primers to recognize and remove specific primer sequences,

supporting samples containing reads amplified by multiple primers.

- Enabled the execution of **Rarefaction** after the Denoising step.
- Added links to **log**, **Summary_Table**, and **Representative_Sequence** within the Portal.
- Modified the **Depth Filter** in Setting.txt to allow filtering by percentage (relative to total reads after Find_unique) in addition to fixed read counts.

■ Version 2.3 (2018/07/09)

- Updated the clustering method (using USEARCH -cluster_smallmem) for sequences with identity below the threshold (defined by UIdentity in Setting.txt). This improves processing speed while maintaining consistent results.
- Fixed a bug where 2_2_Denoise would fail if the sequence count reached zero in 2_1_Find_unique.

■ Version 2.2 (2018/07/01)

- Added support for FASTQ files with primers already removed (Set Primers = No in Setting.txt).
- Added an option to add the read counts of deleted error sequences to the predicted "true" sequences (Set Correct_error = YES in Setting.txt).
- Added **Accession Numbers** (linked to NCBI) to the Detail page accessible from the Portal.
- Added **Start and End times** of the analysis to the Portal.

■ Changes prior to Version 2.2

- Upgraded from **USEARCH v8** to **USEARCH v10**.
- Consolidated all tasks (except phylogenetic tree construction) to run within **USEARCH**.
- Implemented PCR error and chimera removal using the **unoise3** option in USEARCH v10.
- Improved handling of sequences with lower identity (below UIdentity): implemented clustering for sequences assigned the same species name to account for potential multi-species groups.
- Redesigned the **Portal** format.
- Added **log file** output.

【Reference】

- Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJA *et al.* (2016) DADA2: High-resolution sample inference from Illumina amplicon data. *Nature Methods*, **13**, 581-583.
- Edgar RC (2010) Search and clustering orders of magnitude faster than BLAST. *Bioinformatics*, **26**, 2460-2461.
- Frøslev TG, Kjøller R, Bruun HH, Ejrnæs R, Brunbjerg AK *et al.* (2017) Algorithm for post-clustering curation of DNA amplicon data yields reliable biodiversity estimates. *Nature Communications*, **8**, 1188.
- Kumar S, Stecher G, Suleski M, Sanderford M, Sharma S *et al.* (2024) MEGA12: Molecular Evolutionary Genetic Analysis version 12 for adaptive and green computing. *Molecular Biology and Evolution*, **41**, msae263.
- Mahé F, Czech L, Stamatakis A, Quince C, de Vargas C *et al.* (2022) Swarm v3: towards tera-scale amplicon clustering. *Bioinformatics*, **38**, 267-269.
- Rognes T, Flouri T, Nichols B, Quince C, Mahé F (2016) VSEARCH: a versatile open source tool for metagenomics. *PeerJ*, **4**, e2584.

Appendix 1: Details of Setting.txt

- **Red text** (indicated here by bold/code) represents the parameters used in the analysis. Lines starting with # are comments/descriptions.
 - **Do not use spaces** in file names.
 - **Use half-width characters** for all numbers and symbols.
-

◆ Reference Database Settings

DB = [File Name]

Specify the reference file (FASTA format) located in the **DataBase** directory.

Example: DB = MiFish_DB.fas

◆ Primer Settings

Primers = [File Name] or NO

Specify the file containing primer sequences located in the **DataBase** directory. If the primers have already been trimmed from your FASTQ files, set this to "NO".

Example 1: Primers = Primers.txt

Example 2: Primers = NO

MaxDiff = [Value] or length

Specify the maximum number of allowed mismatches in the primer region. Set to "length" to trim the primer region based solely on its length.

Example 1: MaxDiff = 2

Example 2: MaxDiff = length

Shift = [Value]

The number of bases allowed from the start of the read to the 5' end of the primer.

Example: Shift = 4

Divide = YES or NO

Specify whether to split the analysis when reads are amplified by multiple primer pairs. If set to "YES", downstream analyses will be performed separately for each primer.

Example: Divide = YES

◆ Read Length Settings

Length = [Value]

Specify the minimum read length to be used in the analysis.

Example (to use reads $\geq$ 50 bp): Length = 50

◆ Depth Settings (Minimum Frequency)

Depth = [Value] or [Percentage]

Specify the minimum number of identical sequences required for the analysis.

Example 1 (fixed value): Depth = 4

Example 2 (percentage): Depth = 0.01%

Note: If a percentage is specified, the depth is calculated based on the total number of reads per sample after step 2-1 (Find_unique_in_each_sample). For example, at 0.01%, if a sample

has 10,000 reads, Depth = 1; if it has 100,000 reads, Depth = 10.

◆ Denoising Settings

Denoise = YES or NO

Specify whether to perform denoising and use corrected sequences for downstream analysis.

Example: Denoise = YES

Algorithm = 1 or 2

Select the denoising algorithm.

- 1: UNOISE3 (USEARCH/VSEARCH)
- 2: DADA2

Type = 1 or 2 (USEARCH/VSEARCH only)

- 1: Perform denoising for each sample individually.
- 2: Perform denoising after pooling all samples.
- *Note: This item does not need to be removed when using DADA2.*

◆ Rarefaction Settings

Rarefaction = [Value], MIN, or NO

Specify the number of reads for subsampling. Use "MIN" to rarefy to the minimum read count among samples, or "NO" to skip.

Example 1: Rarefaction = 10000

Example 2: Rarefaction = NO

◆ OTU Clustering Settings

Clustering = YES or NO

Specify whether to perform OTU clustering.

Algorithm2 = 1 or 2

- 1: USEARCH/VSEARCH
- 2: SWARM

Cluster_Identity = [Value] (USEARCH/VSEARCH only)

Identity threshold for clustering.

Example (for 97% identity): Cluster_Identity = 97

Difference = [Value] (SWARM only)

The "d" value (local clustering threshold) for SWARM.

Example (d=2): Difference = 2

◆ Post-Clustering Curation

Curation = YES or NO

Specify whether to perform post-clustering curation (e.g., using LULU).

pcid = [Value]

Similarity threshold for curation.

Example: `pcid = 0.84`

◆ Identity Thresholds (Taxonomic Assignment)

UIdentity = [Value]

Upper threshold for taxonomic assignment.

Example (97% or higher): `UIdentity = 97`

LIdentity = [Value]

Lower threshold. Sequences below this value are labeled as "No hit".

Example (below 80% is No hit): `LIdentity = 80`

◆ Dictionary Settings (Located in the Dictionary directory)

Common_name = [File Name] or NO

Japanese common name dictionary. File names must start with "Sname_Jname".

Example: `Common_name = Sname_Jname.togodb.nonredundant.txt`

Family = [File Name] or NO

Family name dictionary.

Example: `Family = Family_name_Fish.txt`

◆ File Management Settings

Temporary = YES or NO

Specify whether to delete temporary FASTQ files generated during preprocessing.

Note: "YES" is highly recommended as these files can become extremely large.

Compress = YES or NO

Specify whether to compress FASTQ files into GZ format.

Note: "NO" is generally recommended because compression is time-consuming and compressed files cannot be analyzed directly by USEARCH.

Appendix 2: Details of Primer.txt

- **Storage:** Save this file in the **DataBase** directory.
 - **Pair Naming:** The names for the Forward and Reverse primers in a pair **must be identical**.
 - **No Spaces:** Do not use spaces in the primer names.
 - **Disabling Primers:** To exclude a primer from the analysis, either delete it or add a # (half-width hash) at the beginning of the line. Using # is recommended if you plan to use the primer again later.
 - **Degenerate Primers:** IUPAC ambiguity codes (degenerate bases) are supported.
 - **Priority:** Primers listed higher in the file have priority. For example, if multiple primer pairs satisfy the mismatch threshold (set by MaxDiff in Setting.txt), the read will be assigned to the pair that appears first in the list.
-

Example Configuration

Forward primers

```
>Primer1
NNNNNNATTTTCGATGTRGTAAGTC
>Primer2
NNNNNNAATCCATGATTCCCGTA
#>Primer3
#NNNNNNCCCGTAGCTTTAAAAGCGC
```

Reverse primers

```
>Primer1
NNNNNNGTATTTACTAGTYAAACC
>Primer2
AAGCTGATGGATGGGAAAT
#>Primer3
#AAAGCGGATCTGAAGTAR
```

*In the example above, **Primer3** will be ignored during the analysis because of the leading #.*

Appendix 3: Demultiplexing with Demultiplex.pl

This script allows for demultiplexing that accounts for the quality scores of index sequences.

1. Generating Raw FASTQ for Demultiplexing

This procedure assumes a PE150 (Paired-End 150bp) run.

1. **Install BCL Convert** on your Linux environment.
(Download: [Illumina Support - BCL Convert](#))
2. **Copy the Run directory** generated by the MiSeq (e.g., 240930_M02728_0074_000000000-DRD3L).
3. **Edit SampleSheet.csv**: Update the following sections and save.
[Reads]
159
159
[Settings]
[Data]
Sample_ID,Sample_Name,Description,I7_Index_ID,index,I5_Index_ID,index2,Sample_Project
Undemulti,Undemulti,,,,,
4. **Edit RunInfo.xml**: Modify the read structure to treat indexes as part of the reads.
Original:
<Read Number="1" NumCycles="151" IsIndexedRead="N" />
<Read Number="2" NumCycles="8" IsIndexedRead="Y" />
<Read Number="3" NumCycles="8" IsIndexedRead="Y" />
<Read Number="4" NumCycles="151" IsIndexedRead="N" />
Modified:
<Read Number="1" NumCycles="159" IsIndexedRead="N" />
<Read Number="2" NumCycles="159" IsIndexedRead="N" />
5. **Run BCL Convert**: Execute the following command within the same directory.
`bcl-convert --bcl-input-directory ./ --output-directory ./out --sample-sheet ./SampleSheet.csv -f`
The generated FASTQ files will be saved in the `out` directory.

2. Executing Demultiplexing

1. **Prepare Files**: Move the output `fastq.gz` files into `PMiFish/Run/non-demultiplexed`.
2. **Create `index.csv`**: Place a file named `index.csv` in the same directory with the following header:
Sample_Name, I7_Index_ID(R1), I5_Index_ID(R2)
3. **Run Script**: Navigate to the directory containing `Demultiplex.pl` and execute:
`perl ./Demultiplex.pl`
Default Settings: Paired-End mode; sequences are discarded if any base in the index has a Q-score < 20.
4. **Options**: You can modify the behavior using the following flags:
-s [10/20/30]: Sets the Q-score threshold (Default: 20).
-t [1/2]: 1 for Single-End, 2 for Paired-End (Default: 2).
5. **Output**: The demultiplexed FASTQ files will be generated in the `PMiFish/Run` directory.

Appendix 4: Phylogenetic Analysis with PA_with_DB.pl

- **Compatibility:** This script has been tested and confirmed to work on **Windows only**.
- **Overview:** This script consolidates reference sequences and taxonomically assigned sequences for each Family to construct **Neighbor-Joining (NJ) trees**.

Prerequisites

1. **Install MEGACC:**
 - Download from the official site: <https://www.megasoftware.net/>
 - Ensure you select the **Command Line (CC)** version and **MEGA12**.
2. **Configuration Files:**
 - Verify that the following setting files are present in the PMiFish/Tools/ directory:
 - muscle_align_nucleotide.mao (for alignment)
 - infer_NJ_nucleotide.mao (for tree construction)

Execution

1. Navigate to the directory containing PA_with_DB.pl.
2. Run the following command:
`perl ./PA_with_DB.pl`

Output

- The analysis results will be generated in the following directory:
PMiFish/Results/5_4_PA_with_DB

If the analysis stops halfway through, it might be due to the files in the "Tools" folder. Please download and install the GUI version of MEGA12, ensure that the button in the bottom right is set to **PROTOTYPE**, and then create and save a new configuration file by following the steps below:

1. **ALIGN** → **MUSCLE DNA Alignment** → Confirm the settings → Save
muscle_align_nucleotide.mao to the **Tools** folder.
2. **PHYLOGENY** → **Construct/Test Neighbor-Joining Tree** → Confirm the settings → Save
infer_NJ_nucleotide.mao to the **Tools** folder.

Appendix 5: Custom Reference Database Creation

While PMiFish was originally developed for **MiFish** (fish eDNA) analysis, it is fully compatible with other taxonomic groups (e.g., crustaceans, fungi) by providing the appropriate reference data.

1. Obtaining and Formatting Reference Data

The reference database must be in FASTA format, with the header following this specific syntax:

```
>gb|[Accession No.]|[Species name]
```

Steps:

1. Search for your target taxonomic group on **NCBI Taxonomy** (<https://www.ncbi.nlm.nih.gov/taxonomy>).
2. Click the **"Nucleotide"** link located under the taxonomic name in the search results.
3. On the left sidebar, filter by **"Mitochondrion"** (if applicable).
 - *Note: If the results are too numerous, refine your search by adding specific gene names (e.g., 12S, COI).*
4. Use the **"Send to"** menu, select **"File"**, choose **"GenBank"** format, and download the data.
5. Place the downloaded sequence.gb file into the PMiFish/Database directory.
6. Execute the following command in that directory:

```
perl ./GB2DB.pl
```
7. The formatted reference file, database.fas, will be generated in the same directory.

2. Extracting Amplified Regions (In-silico PCR)

To extract the specific regions amplified by your primers:

1. Execute the following command in the directory containing database.fas:

```
perl ./Amplified_region.pl -in database.fas -f [forward_primer] -r [reverse_primer]
```

 - *Example:* perl ./Amplified_region.pl -in database.fas -f ATGCATCG... -r GGGGSTGST...
 - *Note: Degenerate base notations are supported.*
2. The extracted sequences will be saved as amplified_region.fas in the same directory.

3. Retrieving Taxonomic Ranks from NCBI

To obtain the full taxonomic hierarchy (Kingdom, Phylum, Class, Order, Family, Genus, Species):

1. Download taxdump.tar.gz from the NCBI FTP site:
<https://ftp.ncbi.nih.gov/pub/taxonomy/>
2. Extract the archive and place the names.dmp and nodes.dmp files into the PMiFish/Database directory.
3. Execute the following command:

```
perl ./Taxonomic_Rank.pl [Taxon_Name]
```

 - *Example:* perl ./Taxonomic_Rank.pl "Mammalia"
 - *Note: While the output includes seven primary ranks, you can search using sub-ranks (e.g., Suborder, Subfamily).*
4. The results will be output to Taxonomic_Ranks.tsv.

5. **Creating a Family Dictionary:** You can create a Family-level dictionary by extracting only the "Family" and "Genus" columns from the output, saving them as a text file, and placing it in the PMiFish/Dictionary directory.

Appendix 6: Clustering Reference Sequences

- **Background:** Reference databases often contain sequences labeled with the same species name that are genetically distinct.
- **Purpose:** By grouping sequences of the same species and clustering them based on a defined identity threshold, intraspecific clusters can be accurately reflected during taxonomic assignment.

Execution

1. Navigate to the PMiFish/Database directory.
2. Run the following command:

```
perl ./Cluster_maker.pl -in [database_name] -id [cluster_identity]
```

 - *Example:* perl ./Cluster_maker.pl -in database.fas -id 98.5

Output

- A new file named [database_name]_clusterized.fas will be generated in the same directory.
- If clusters are identified based on your threshold, the headers will be modified as follows:
 - >gb|[Accession No.]|Species_A_cluster1
 - >gb|[Accession No.]|Species_A_cluster1
 - >gb|[Accession No.]|Species_A_cluster2
- *Note: The total number of sequences remains the same as in the original database.fas (sequences are not merged; only the headers are updated with cluster IDs).*

Usage

- Use the generated _clusterized.fas file as your reference database for subsequent PMiFish analyses.

Appendix 7: Troubleshooting

■ The script does not run on Windows

If you are attempting to run a **32-bit version of USEARCH** on a **64-bit Windows PC**, you may occasionally encounter an error message such as "**Can't spawn...**" and the process will fail to start.

Solution: This issue is often caused by missing runtime components. Please download and install the "**Visual C++ Redistributable for Visual Studio 2015**" from the official Microsoft website:

- **Download URL:** <https://www.microsoft.com/en-us/download/details.aspx?id=48145>